INCH-POUND

QQ-S-763F
June 27, 1996
SUPERSEDING
QQ-S-763E
July 31, 1986

FEDERAL SPECIFICATION

STEEL BARS, WIRE, SHAPES, AND FORGINGS, CORROSION RESISTANT

The General Services Administration has authorized the use of this federal specification by all federal agencies

1 SCOPE AND CLASSIFICATION

1 1 Scope This specification covers corrosion-resistant steel bars, wire, shapes, and forgings

1 2 Classification

1 2 1 Corrosion-resistant steel shall be furnished in the following classes, conditions, forms, and finishes, as specified (see 6 2)

1 2 1 1 Classes and conditions The material shall be furnished in the classes and conditions shown in Table I

1 2 1 2 Forms

1 2 1 2 1 Bars Any size that is round, square, rectangular (excluding plate), hexagon, or octagon, furnished in straight lengths regardless of finish, that is produced by rolling, extruding, forging, etc Flat rolled product up to 10 inches inclusive in width and 0 125 inches and over in thickness is classified as bar (All cold reduced flat material under 0 1875 inches in thickness is classified as strip if it is over 0 375 inches in width)

1 2 1 2 2 Wire Any size round or shaped cold finished product that is supplied in coils

1 2 1 2 3 Forgings Parts produced by hot mechanical shaping of such products as bars, billets or other semifinished materials, using hammers presses and forging machines

Beneficial comments, recommendations, additions, deletions, clarifications, etc , and any other data which may improve this document should be sent to Commander, Naval Air Warfare Center Aircraft Division, Code 4 1 4 2B120-3, Highway 547, Lakehurst, NJ 08733-5100

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1 2 1 2 4 Shapes A solid section other than bar or wire, furnished in straight lengths (includes structural angles, channels, tees, and zees)

1 2 1 3 Finish

1 2 1 3 1 Hot finished (conditions A, T, and H only, bar only) Bars in the hot-finished condition can be ordered with one of the following finishes

a Hot finished, scale not removed (excluding spot conditioning)
b Pickled or blast cleaned and pickled
c Rough turned (round bars only)

1 2 1 3 2 Cold finished (bars only) Bars in the cold-finished condition can be ordered with one of the following finishes

a Cold drawn or cold rolled (conditions A and B only)
b Centerless ground or smooth turned (all conditions, round bars only)
c Polished (all conditions, round bars only)

1 2 1 3 3 Wire Wire is usually furnished with a cold drawn finish, any exceptions need to specified in the purchase order

2 APPLICABLE DOCUMENTS

2 1 Government publications The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

Federal Standards

Fed Std No 123	-	Marking for Domestic Shipment (Civilian Agencies)
Fed Std No 183	-	Continuous Identification Marking of Iron and Steel Products

(Activities outside the Federal Government may obtain copies of federal specifications, standards, and commercial item descriptions, as specified in the General Information section of the Index of Federal Specifications, Standards and Commercial Item Descriptions The Index is for sale on a subscription basis from the Superintendent of Documents, U S Government Printing Office, Washington, DC 20402

(Single copies of this specification and other federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407)

(Federal Government activities may obtain copies of Federal

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standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies)

Military Specifications

MIL-H-6875 - Heat Treatment of Steel, Process for

Military Standards

MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2 2 Other publications The following documents form a part of this specification to the extent specified herein Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply

American Society for Testing and Materials (ASTM)

ASTM A 262	-	Detecting Susceptibility to Intergranular Attack in Austenitic Stainless Steels
ASTM A 342	-	Permeablility of Feebly Magnetic Materials
ASTM A 370	-	Mechanical Testing of Steel Products
ASTM A 484/ 484M	-	General Requirements for Stainless and Heat-Resisting, Bars, Billets and Forgings,
ASTM A 555	-	General Requirements for Stainless and Heat-Resisting Steel Wire and Wire Rods
ASTM A 751	-	Standards Test Methods, Practices, and Terminolgy for the Chemical Analysis of Steel Products
ASTM D 3951	-	Commercial Packaging

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society for Testing and Materials, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue,

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Philadelphia, PA 19111-5094)

2 3 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

3 REQUIREMENTS

3 1 Material The material may be made by one or more of the following processes electric furnace, electric-induction, vacuum furnace or other suitable commercial processes If a specific melting practice is required by the purchaser, it shall be specified on the purchase order (see 6 2)

3 2 Non-finishing surface processing

3 2 1 Grinding Bars and forgings may be ground to remove surface defects, provided such grinding does not reduce the thickness or width at any point below the allowable dimensional tolerances An abrasive wheel shall be used for such grinding and shall be operated at a speed proper to insure that defective areas are cleanly cut out Grinding shall not be so severe as to change the metallurgical condition of the material

3 2 2 Cleaning Unless otherwise specified in the contract or order, structural shapes shall be subjected to a final cleaning treatment for the removal of scale, by the use of an appropriate cleaning solution such as nitric or nitric and hydrofluoric acid (see 6 2)

3 3 Rough forgings Rough forgings shall have sufficient excess stock to permit finishing to required dimensions without excessive waste Allowances for machining shall be specified (see 6 2)

3 4 Chemical composition The material shall conform to the chemical composition shown in Table II and shall be within the check analysis tolerances shown in ASTM A 484 or ASTM A 555 as applicable

3 5 Mechanical properties The material shall conform to the mechanical properties shown in Table III for the respective conditions Heat treatable grades (400 series) shall develop the properties specified for the T or H conditions when subjected to thermal treatment in accordance with MIL-H-6875 and as specified or recommended by the purchaser

3 6 Macrostructure For classes 410, 420, 440A, 440B, and 440C, the macrostructure of the material shall be sound, free from pipes, fissures, gas cavities, sponginess, abnormal inclusions or segregations, or unusually numerous pin-holes when tested in accordance with 4 5 3 This requirement applies only to the classes specified in this paragraph

3 6 1 Forging Grain Flow The selection of forging blank size and orientation and forging technique shall provide a grain flow pattern essentially parallel to major-stressed surface areas of the finished part as

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indicated by design information (see 6 2) The grain flow pattern shall be free from re-entrant and sharply folded flow lines

3 7 Magnetic permeability When low magnetic permeability is specified (see 6 2), class 304, in the annealed condition, shall show a magnetic permeability not higher than 1 02 at 200 oersteds (air equal to 1 00) When low magnetic permeability is specified for classes other than 304, the acceptable magnetic permeability values shall be as specified (see 6 2) and the inspection procedure shall be in accordance with ASTM A 342

3 8 Resistance to intergranular corrosion (precipitated carbides) Classes 304 - condition A, 304L, 316 - condition A, 316L, 317 - condition A, 321, and 347 shall be free from precipitated grain boundary carbide networks which result in intergranular corrosion These steels shall be considered acceptable when specimens pass the specified test of 4 5 4 This requirement applies only to the classes and conditions specified in this paragraph

3 9 Dimensional tolerances

3 9 1 Bars, shapes, and wire For bars and shapes, the tolerances shall conform to the applicable requirements of ASTM A 484 For wires, the tolerances shall conform to the applicable requirements of ASTM A 555 When exact lengths are not ordered (see 6 2), bars will be accepted in mill lengths of 6 to 20 feet but not more than 10 percent of any order shall be furnished in lengths shorter than 10 feet

3 9 2 Structural shapes

3 9 2 1 Weight Structural shapes of 6 pounds per linear-foot or less will be acceptable if the actual weights are not over 7-1/2 percent above or 7-1/2 percent below the ordered weights Shapes over 6 pounds per linear foot will be accepted if the weights are not over 4-1/2 percent above or 4-1/2 percent below the ordered weights

3 9 2 2 Size Sections having legs or flanges up to 6 inches, inclusive, shall not exceed 1/8 inch over or under the ordered length or width of legs or flanges Sections having legs or flanges over 6 inches shall not exceed 3/16 inch over and 1/8 inch under the ordered length or width of legs and flanges The maximum depth of grinding for spot conditioning shall not exceed 10 percent of the thickness of the shape at any point of conditioning

3 9 3 Forgings All forgings shall conform to the sizes and shapes as specified (see 6 2) When dimensional tolerances are not included in the contract or order, forgings measured on their diameters or between parallel faces shall not vary from the specified dimensions by more than plus 3/32 inch on smooth forgings or plus 1/32 inch on rough machined forgings Dimensional tolerances for finished forgings shall be as specified on the applicable drawings

3 10 Identification marking

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3 10 1 Bars (continuous marking) When specified by the procuring activity, continuous identification marking shall be in accordance with Fed Std No 183 Each round, square, and flat bar 1 inch and over, each hexagon, square, and flat 7/8 inch and over, and each octagon 1 inch and over shall be printed in ink with constantly recurring symbols including a coding of the name or trademark of the manufacturer and an identifying designation consisting of the class number and a condition designator selected from Fed Std No 183 The symbols shall be repeated at intervals not greater than 3 feet The identifying designator shall be coded as shown in the following example (see 6 2)

Class number 304
Condition designator A

For rounds, and bars less than 1 inch, hexagons, squares and flats less than 7/8 inch, and octagons less than 1 inch, the same information shall be marked or printed on substantial tags securely affixed to each end of each bundle with a third tag placed inside the bundle near the middle

3 10 2 Shapes and shaft forgings Shapes and shaft forgings shall be die or rubber stamped or marked with a non-water-soluble ink on one end with the following information

Class number
Condition designator

3 10 3 Forgings (except shaft forgings) Each forging shall be marked with the manufacturer's name or trademark, drawing or die number, heat and forging number, class number, and a condition designator Small forgings shipped in bundles, of such size that individual marking is not practical, shall have the above information printed on substantial tags securely affixed to each end of each bundle with a third tag placed inside the bundle near the middle

3 10 4 Wire Unless other identification is specified in the contract or purchase order (see 6 2), wire shall be identified by metal tags, impression stamped with the legend specified in FED-STD-183, and shall be attached to each coil

3 11 Workmanship The material shall be uniform in quality and condition, free from pipe and flakes or heat checks, and shall contain no welds or defects such as seams, laps, cracks, slag, hard spots, porosity, slivers, scabs, rolled-in scale, fissures, gas cavities, sponginess, nonmetallic inclusions, and undue segregation, which due to the nature, degree, or extent may detrimentally affect the suitability for the service intended

3 12 Heat Treatment Heat treatment shall be in accordance with MIL-H-6875 AMS 2750 pyrometry requirements do not apply to furnaces used only for heat treating material to condition A The annealing procedure for S30430 and

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384 shall be in accordance with the table below, in addition to the requirements set forth in MIL-H-6875

AISI or UNS Designation	Annealing treatment	
	Heating °F	Cooling
S30430	1850 to 2100	Water quench
384	1900 to 2100	Water quench

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements

4 2 Lot

a For the purpose of chemical analysis a lot shall consist of all bars, wire, shapes, and forgings made from the same heat and presented for inspection at the same time

b For all other tests, a lot shall consist of all product of the same size, same heat, and produced under the same processing conditions For austenitic, ferritic, and martensitic classes in the annealed condition, a lot may consist of product that is heat treated in more than one charge in the same furnace, or in several furnaces, using the same processing parameters (see 6 7) For martensitic stainless steels heat treated to the T or H condition, however, a lot shall consist of product of the same size, same heat, and the same heat treat charge in a batch type furnace, or under the same processing conditions without interruption within 24 hours in a continuous type furnace

4 2 1 Forgings Unless another lot definition is specified in the contract or order (see 6 2), a lot shall consist of one forging for forgings weighing 250 pounds or more, and 1000 pounds maximum for forgings weighing less than 250 pounds each with the forgings in each lot to be of one class and shape only, made from the same heat and subjected to the same heat treating procedure (annealing or tempering)

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4 3 Sampling procedure

4 3 1 Chemical analysis For the purpose of chemical analysis of 4 5 1, one sample shall be taken from each lot When a lot is a portion of a heat previously subjected to chemical analysis and in compliance with the specified composition range, additional analysis sampling is waived In the event a portion of a lot is not properly identified, an analysis shall be made of each piece or as required by the purchaser Sampling shall conform to the applicable chemical analysis document specified in ASTM A 751

4 3 2 Macrostructure examination Samples shall be selected as shown below In special cases where more than the usual number of macrostructure etch tests are required, the number of such tests shall be as specified in the contract or order (see 6 2)

4 3 2 1 Heat Qualification

4 3 2 1 1 Heats of top poured ingots Samples shall be taken from semi-finished or finished product representing the top and bottom of the first ingot and last usable ingot from heats having not more than 10 ingots or not over 30 tons or from portions of heats within these limits, and from the top and bottom of the first, middle, and last usable ingot of heats having more than 10 ingots or over 30 tons

4 3 2 1 2 Heats of bottom poured ingots Samples shall be taken from semi-finished or finished product representing the top and bottom of three ingots One ingot shall be taken at random from the first usable plate poured, one ingot at random from the usable plate poured nearest to the middle of the heat, and one ingot at random from the last usable plate poured When a heat consists of two usable plates, two of the sample ingots shall be selected from the second usable plate poured When a heat consists of a single usable plate, any three ingots at random may be selected If there are less than three ingots in the heat, samples shall be taken representing the top and bottom of all ingots

4 3 2 1 3 Strand cast heats Samples shall be taken from semi-finished or finished product having at least a 3 1 reduction in cross-section from the cast strand, or samples of the as-cast strand similarly reduced, representing the front, middle, and back of both strands when two strands are cast, or of an inside strand and an outside strand when more than two strands are cast When a single strand is cast, six samples having at least a 3 1 reduction from the cast strand, or samples of the cast strand similarly reduced, representing both ends of the first, middle, and last usable cuts (blooms) of the strand or product shall be taken

4 3 2 2 Product qualification Samples shall be taken at random from not less than 10 percent of the pieces of each lot A lot shall be all product of one size from one heat in one shipment Not less than three nor more than ten samples shall be selected from a lot, except that if the quantity in the lot is three pieces or less, one sample shall be taken from each piece

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4 3 3 Mechanical tests Sample(s) from each lot of bars, wire and shapes (rolled, extruded or forged) and from each lot of forgings, shall be selected for the tests specified in 4 5 2 Test specimens shall be taken from the selected pieces as follows

4 3 3 1 Bars, wire, and shapes One longitudinal tension test specimen shall be taken from each sample item and tested as specified in 4 5 2

4 3 3 2 Forgings

4 3 3 2 1 250 pounds and over Unless otherwise specified in the contract or order, each forging weighing 250 pounds or more shall be tested individually One tension test specimen shall be taken from a prolongation on each end of each forging The test specimens shall be taken midway between the center and outside of the cross section of the forging

4 3 3 2 2 Under 250 pounds For forgings weighing less than 250 pounds two tension test specimens shall be taken from two sample forgings to represent the lot One tension test specimen shall be taken from one forging and the remaining tension test specimen from the other forging Test specimens shall be taken from suitable prolongations of the forgings, as specified by the procuring activity (see 6 2), or at the option of the contractor, forgings in excess of the number required may be provided for tests For each forging weighing less than 250 pounds submitted individually, one tension specimen shall be taken

4 3 3 2 3 Multiples Where forgings are made in multiples from a single forging, that is, forged in one piece and machined apart, individual tests of each forging need not be made, but tests of the large forging shall govern

4 3 4 Magnetic permeability When specified (see 3 7), one sample shall be selected from each lot The specimens shall be 3 inches long, the thickness or diameter shall be 1/2 inch for material 1/2 inch and over in thickness or diameter, and the full size of the material, if the material is under 1/2 inch in thickness or diameter The specimens shall be ground and polished all over Specimens, since they represent annealed material, may be annealed and pickled after machining

4 3 5 Precipitated carbides test A minimum of one sample shall be selected from each lot for the test specified in 4 5 4 to determine conformance to the requirements of 3 8

4 4 Examination

4 4 1 Dimensional and visual surface inspection All material shall be subject to dimensional and visual surface inspection to determine whether the material conforms to this specification Lots containing defective material shall be rejected

4 4 2 Reinspection Lots rejected on account of dimensional or visual

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surface defects may be resubmitted for inspection in accordance with 4 4 1 after the manufacturer has reworked and reinspected them to remove nonconforming material

4 5 Tests

4 5 1 Chemical analysis Samples selected in accordance with 4 3 1 shall be analyzed to determine conformance with Table II in accordance with any one of the chemical analysis methods specified in ASTM A 751 If any sample fails to conform to Table II, the entire lot shall be rejected

4 5 2 Mechanical tests

4 5 2 1 Tensile test The tension tests shall be made in accordance with ASTM A 370 using either the 0 2% offset method or the extension under the load method using the limiting plastic strain of 0 002 in/in when determining the yield strength The tensile specimens shall be prepared in accordance with ASTM A 370, form and dimension as applicable No lot will be accepted if the yield strength of any one specimen is below the minimum yield strength shown in Table III

4 5 2 2 Hardness test The number of hardness tests shall be sufficient to establish the uniformity of hardness of the material in each lot Hardness tests, as applicable for specimen size, shall be conducted in accordance with ASTM A370 Failure of any sample to meet the requirements of this specification shall be cause for rejection of the lot

4 5 3 Macrostructure test The specimens for the macrostructure test shall be cut from the ends of the selected sample and shall represent the full cross section bar, billet, wire, shape, or forging The surfaces of the specimens to be examined shall be prepared for etch testing The prepared specimens shall first be cleaned and heated in water to the same temperature as the acid etching solution It shall then be immersed in a solution consisting of equal parts, by volume, of concentrated hydrochloric acid and water at approximately 160°F , for a period of time sufficient to develop fully the metallurgical structure Fresh acid shall be used for each lot of specimens After etching, the specimens shall be washed in running water or steam and any deposit removed by scrubbing The dried specimen shall then be dipped in cold concentrated nitric acid, washed in cold water, and dried If the specimen fails to conform to 3 6, the lot represented shall be rejected

4 5 4 Intergranular corrosion tests (precipitated carbides) Specimens selected as specified in 4 3 5 shall be tested in accordance with Practice E of ASTM A 262

4 5 5 Rejection and retest Failure of a specimen to meet a test requirement shall be cause for rejection of the lot At the discretion of the procuring activity, retest will be permitted (see 6 2) A retest sample of three specimens shall be tested to replace each failed specimen of the

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original sample If one of the retest specimens fails, the lot shall be rejected with no further retesting permitted Where failure of any lot of material to meet the requirements to this specification is due to inadequate heat treatment, the material may be reheat-treated and resubmitted for test Only two such reheat-treatments shall be allowed

4 6 Inspection of preparation for delivery The preservation, packing, and marking of the bars, shapes, wire and forgings shall be examined to determine compliance with the requirements of section 5

5 PACKAGING

5 1 Preservation Preservation is not required

5 2 Packing Packing shall be Level A or Level C, as specified

5 2 1 Level A Bars, shapes and wire shall be prepared for shipment in accordance with MIL-STD-163 Forgings of such size requiring consolidation or having fragile appendages shall be packed in containers specified in MIL-STD-163 Polished surfaces shall be interleaved or otherwise protected with a nonabrasive paper

5 2 2 Level C The subject material shall be segregated by class, heat, condition, form, size, and finish as applicable, prior to packing Packing of the material shall be in accordance with ASTM D 3951 In addition, polished surfaces shall be interleaved or otherwise protected with a nonabrasive paper

5 3 Marking

5 3 1 Military agencies In addition to any special marking required in the contract or order (see 6 2), marking for shipment shall be in accordance with MIL-STD-163

5 3 2 Civil agencies In addition to any special marking required in the contract or order (see 6 2), marking of shipping containers shall be in accordance with Fed Std No 123 Unless otherwise specified in the contract or order, shipments shall be marked with the name of the material, the class and condition, and the quantity contained therein as defined by the contract or order under which shipment is made, the name of the contractor, the number of the contract or order, and the gross weight

6 NOTES

INFORMATION FOR GUIDANCE ONLY This section contains information of a general or explanatory nature which is helpful, but is not mandatory

6 1 Intended use

6 1 1 Classes 202 302 and 305 Where sensitization cannot be tolerated

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classes 202, 302 and 305 are not recommended for use

6 1 1 1 Classes 202 and 302 Class 302 is intended for use where severe corrosion is not a problem and no welding other than spotwelding is employed except when welding is followed by annealing Class 202 is intended as a substitute for class 302

6 1 1 2 Class 305, 384, and UNS S30430 These materials are intended for use where a lower rate of work hardening than classes 302 or 304 and less change of magnetic permeability are required UNS S30430 is primarily used for cold headed fasteners

6 1 2 Class 304, condition B This material is intended primarily for structural applications where welding is limited to spot welding unless welding is followed by annealing

6 1 3 Classes 304L, 316L, 321, 347, 304, and 316 - condition A These materials are generally preferred for resistance to most severely corrosive media, and are generally more resistant than the other types

6 1 4 Class 317, condition A This material exhibits superior corrosion resistance to most severely corrosive media and is used in special applications

6 1 5 Classes 316, 317 When these Molybdenum-bearing stainless steels are intended for use in nitric acid environments, corrosion testing in accordance with ASTM A 262, practice C is recommended

6 1 6 Classes 304L, 316L, 321 and 347 - condition A These materials are intended primarily for use in applications where welding is necessary, subsequent annealing and quenching is impracticable, and exposure to most severely corrosive media is involved

6 1 7 Classes 309 and 310 These materials are intended for the highest temperature applications requiring high temperature strength and resistance to oxidation

6 1 8 Class 403 This material is intended for use in applications where high mechanical properties, corrosion, abrasive wear and wet erosion resistance are required

6 1 9 Class 405 This material is used in applications where marked hardening is undesirable when cooling from elevated temperatures

6 1 10 Classes 410 and 430 These materials are intended for structural parts in application where corrosive conditions are not severe, class 430 being the better alloy Class 410 is hardenable by heat treatment, class 430 is not hardenable These alloys are subject to rusting and pitting when exposed to sea water and will rust in salt air

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6 1 11 Class 414 This material is heat treatable to slightly higher mechanical properties than those of class 410 Corrosion resistance is similar to class 410

6 1 12 Class 420 This material is intended for use in ball and roller bearings, cutlery, and other parts requiring high hardness

6 1 13 Class 440A This material is used where material of greater hardness than class 420 and greater toughness than class 440B and 440C is required

6 1 14 Class 440B This material is used for applicatons requiring an intermediate hardness between class 440A and 440C

6 1 15 Class 440C This material is suitable for use where extremely high hardness and wear-resistance are desirable in a corrosion-resistant steel It is intended for use in ball and roller bearings and races, cutting edges, shear blades, surgical and dental equipment, valve seats and other applications requiring high hardness

6 1 16 Class 446 This material is used principally for the manufacture of parts which must resist high temperatures in service without scaling

6 2 Ordering data Purchasers should select the preferred options permitted herein and include the following information in procurement documents

(a) Title, number, and date of this specification
(b) Class, condition, form and finish (1 2 and Table I)
(c) Specify melting practice, if required (see 3 1)
(d) Specify other cleaning processes or if cleaning is not required (see 3 2 2)
(e) Allowances for machining (3 3)
(f) Major stressed surface areas of the finished part (see 3 6 1)
(g) Whether low magnetic permeability is required for class 304 condition A, and the acceptable permeability values for classes other than 304 if such testing is desired (see 3 7)
(h) Whether exact or mill lengths are required (see 3 9 1)
(i) Size and shape of forgings (see 3 9 3)
(j) Whether continuous marking or other identification marking is required (see 3 10 1 and 3 10 4)
(k) Other lot definition for forgings (see 4 2 1)
(l) Number of macrostructure samples required for special cases (see 4 3 2)
(m) Forging prolongations for testing (see 4 3 3 2 2)
(n) Whether retests are permitted (see 4 5 5)
(o) Whether the material should be prepared for shipment by level A or C (see 5 2)
(p) Whether special marking for shipment is required (see 5 3)

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6 3 Class 431 Class 431 under QQ-S-763C should be purchased to MIL-S-18732

6 4 Free machining grades Classes 303, 303Se, 416, 416Se, 430F and 430FSe under QQ-S-763C should be purchased to ASTM A 582 or ASTM A 581

6 5 Cross-reference of designations Designations of the various classes covered by this specification and the designations of the most nearly equivalent classes in other Government specifications are shown in Table IV This table is for reference purposes only

6 6 Subject term (key word) listing

Corrosion-resistant steel
Forgings, steel
Shapes, steel
Steel bars
Steel wire

6 7 Equivalent processing parameters Product of the same size and from the same heat that is heat treated in different furnace charges can be considered to be a part of the same lot provided the heat treating procedure is the same and all furnaces used are similar in size and meet the temperature uniformity and accuracy requirements of a documented furnace quality assurance program

6 8 Changes from previous issue Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

MILITARY INTEREST

Custodians

Army-MR
Navy-AS
Air Force-11

Review Activities

Army-MI, AT, ME, AR
Navy-SH, OS, YD
DLA-IS

CIVIL AGENCY COORDINATING ACTIVITIES

GSA-FSS
NASA
DOE

PREPARING ACTIVITY

NAVY-AS

DOD Project 9510-0941

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Table I Classes and conditions

Condition 1/				
Class	Annealed "A"	Cold Worked High Tensile "B"	Inter-mediate temper " T "	Hard Temper " H "
202	A	B	-	-
302	A	B	-	--
UNS S30430	A	-	-	-
304	A	B	-	-
304L	A	-	-	-
305	A	-	-	-
309	A	-	-	-
310	A	-	-	-
316	A	B	-	-
316L	A	-	-	-
317	A	B	-	-
321	A	-	-	-
347	A	-	-	-
384	A	-	-	-
403	A	-	T	H
405	A	-	-	-
410	A	-	T	H
414	A	-	-	H
420	A	-	-	-
430	A	-	-	-
440A	A	-	-	-
440B	A	-	-	-
440C	A	-	-	-
446	A	-	-	-

1/ Forgings (conditions A, T and H only)

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Table II Chemical composition, percent 1/

Class	Carbon	Manganese	Phosphorous	Sulphur	Silicon	Chromium	Nickel	Molybdenum	Titanium	Columbium + Tanatalum	Aluminum	Nitrogen	Copper
202	0 15	7 50/ 10 00	0 06	0 03	1 00	17 0/ 19 0	4 0/ 6 0	---	---	---	---	0 25	---
302	0 15	2 00	0 045	0 03	1 00	17 0/ 19 0	8 0/ 10 0	1 00	---	---	---	0.10	1 00
304	0 08	2 00	0 045	0.03	1 00	18 0/ 20 0	8 0/ 10 5	1 00	---	---	---	0 10	1 00
304L	0 03	2 00	0 045	0 03	1 00	18 0/ 20 0	8.0/ 12 0	1 00	---	---	---	0 10	1 00
S30430	0 03	2 00	0 045	0 03	1 00	17.0/ 19 0	8 0/ 10 0	---	---	---	---	---	3 0/ 4 0
305	0 12	2 00	0 045	0.03	1.00	17 0/ 19 0	10 50/ 13 0	1 00	---	---	---	---	1.00
309	0.20	2.00	0.045	0.03	1.00	22 0/ 24 0	12 0/ 15 0	1 00	---	---	---	---	1 00
310	0 25	2 00	0 045	0.03	1 50	24 0/ 26.0	19 0/ 22 0	1.00	---	---	---	---	1.00
316	0 08	2 00	0 045	0.03	1 00	16 0/ 18 0	10 0/ 14 0	2 0/ 3 0	---	---	---	0 10	1 00
316L	0 03	2 00	0 045	0 03	1 00	16 0/ 18 0	10 0/ 14 0	2 0/ 3 0	---	---	---	0 10	1 00
317	0 08	2 00	0 045	0 03	1 00	18 0/ 20 0	11 0/ 15 0	3 0/ 4.0	---	---	---	0 10	1 00
321	0 08	2 00	0 045	0 03	1 00	17 0/ 19 0	9 0/ 12 0	1 00	5X carbon min	---	---	---	1.00

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Table II. Chemical composition, percent 1/ (continued)

Class	Carbon	Manganese	Phosphorous	Sulphur	Silicon	Chromium	Nickel	Molybdenum	Titanium	Columbium + Tantalum	Aluminum	Nitrogen	Copper
347	0 08	2 00	0 045	0 03	1 00	17 0/ 19 0	9 0/ 13 0	1.00	---	10X carbon min	---	---	1 00
384	0.04	2 00	0 045	0 03	1 00	15 0/ 17 0	17 0/ 19.0	---	---	---	---	---	1 00
403	0 15	1.00	0 040	0.03	0 50	11 5/ 13 0	---	---	---	---	---	---	---
405	0.08	1 00	0 040	0.03	1.00	11 5/ 14 5	---	---	---	---	0.10/ 0 30	---	---
410	0.15	1 00	0.040	0 03	1.00	11 5/ 13.5	---	---	---	---	---	---	---
414	0.15	1 00	0 040	0 03	1.00	11 5/ 13 5	1 25/ 2 50	---	---	---	---	---	---
420	0.15 min	1 00	0 040	0 03	1 00	12.0/ 14 0	---	---	---	---	---	---	---
430	0 12	1.00	0 040	0 03	1 00	16 0/ 18 0	---	---	---	---	---	---	---
440A	0 60/ 0.75	1 00	0 04	030	1 00	16 0/ 18 0	---	0 75	---	---	---	---	---
440B	0 75/ 0.95	1 00	0 04	.030	1 00	16 0/ 18.0	---	0 75	---	---	---	---	---
440C	0 95/ 1 20	1 00	0 04	030	1 00	16 0/ 18 0	---	0 75	---	---	---	---	---
446	0.20	1 50	0 04	.030	1.00	23.0/ 27 0	---	---	---	---	---	0 25	---

1/ Single values are maximum values unless otherwise indicated

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Table III Mechanical properties

Classes	Condition	Finish	Diameter or Thickness inches	Yield strength (min) 0 2 percent offset psi	Ultimate tensile strength (min)	Elongation in 2 inches (min) percent	reduction in area (min) percent	Brinnell hardness (max) 4/
202, 302, 304, 305 309, 310, 316 317, 321, 347	A	Hot	0 500 and less	-----	115,000 (max)	----	----	----
			over 0 500	30,000	75,000 1/	40	50	----
		Cold	0 500 and less	-----	125,000 (max)	---	----	----
			over 0 500	30,000	75,000 1/	30	50	----
202, 302, 304	B	Cold	0 500 and less	---	125/155,000	----	----	----
			over 0 500 to 0 750	100,000	125,000	12	35	----
			0 751 to 1 000	80,000	115,000	15	35	----
			1 001 to 1 250	65,000	105,000	20	35	----
			1 251 to 1 500	50,000	100,000	28	45	----
			1 501 to 1 750	45,000	95,000	30	45	----
			over 1 750	30,000	75,000	35	50	----
316, 317	B	Cold	0 500 and less	----	110/140,000	----	----	----
			0 501 to 0 750	95,000	110,000	15	45	----
			0 751 to 1 000	80,000	100,000	20	45	----
			1 001 to 1 250	65,000	95,000	25	45	----
			1 251 to 1 500	50,000	90,000	30	45	----
304L 316L	A	Hot	0 500 and less	----	115,000 (max)	----	----	----
			over 0.500	25,000	70,000	40	50	----
		Cold	0 500 and less	----	115,000 (max)	----	----	----
			over 0 500	25,000	70,000	30	40	----

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Table III Mechanical Properties (continued)

Classes	Condition	Finish	Diameter or Thickness inches	Yield strength (min) 0 2 percent offset psi	Ultimate tensile strength (min)	Elongation in 2 inches (min) percent	reduction in area (min) percent	Brinnell hardness (max) 4/
403, 410	A	Hot	0 500 and less	----	115,000 (max)	----	----	----
			over 0.500	----	----	----	----	241
		Cold	0 500 and less	----	115,000 (max.)	----	----	----
			over 0.500	----	----	----	----	241
	T	Hot	0 500 and less	----	100/130,000	----	----	----
			over 0.500	80,000	100,000	15	45	----
		Cold	0 500 and less	----	100/130,000	----	----	----
			over 0.500	80,000	100,000	12	40	----
	H	Hot	0.500 and less	----	120/150,000	----	----	----
			over 0 500	90,000	120,000	12	40	----
		Cold	0.500 and less	----	120/150,000	----	----	----
			over 0.500	90,000	120,000	12	40	----
405 2/	A	Hot	0 500 and less	----	100,000 (max.)	----	----	----
			over 0 500	----	----	----	----	207
		Cold	0 500 and less	----	105,000 (max)	----	----	----
			over 0.500	----	----	----	----	217
414	A	Hot	0 500 and less	----	135,000 (max)	----	----	----
			over 0 500	----	----	----	----	286
		Cold	0 500 and less		135,000 (max)	----	----	----
			over 0 500	----	----	----	----	286

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Table III Mechanical Properties (continued)

Classes	Condition	Finish	Diameter or Thickness inches	Yield strength (min) 0 2 percent offset psi	Ultimate tensile strength (min)	Elongation in 2 inches (min) percent	reduction in area (min) percent	Brinnell hardness (max) 4/
414	H	Hot	0 500 and less	----	120/150,000	----	----	----
			over 0 500	100,000	120,000	15	45	----
		Cold	0 500 and less	----	120/150,000	----	----	----
			over 0 500	100,000	120,000	12	40	----
420 3/	A	Hot	0.500 and less	----	115,000 (max)	----	----	----
			over 0 500	----	----	----	----	241
		Cold	0 500 and less	----	115,000 (max)	----	----	----
			over 0 500	----	----	----	----	241
430	A	Hot	0 500 and less	----	100,000 (max)	----	----	----
			over 0.500	35,000	65,000	20	50	----
		Cold	0 500 and less	----	110,000 (max)	----	----	----
			over 0 500	35,000	65,000	15	45	----
440A 3/	A	Hot	0 500 and less	----	115,000 (max)	----	----	----
			over 0 500	----	----	----	----	241
		Cold	0 500 and less	----	120,000 (max)	----	----	----
			over 0 500	----	----	----	----	255
440B 3/	A	Hot	0 500 and less	----	125,000 (max)	----	----	----
			over 0 500	----	----	----	----	255
		Cold	0 500 and less	----	130,000 (max)	----	----	----
			over 0 500	----	----	----	----	269

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Table III Mechanical Properties (continued)

Classes	Condition	Finish	Diameter or Thickness inches	Yield strength (min) 0 2 percent offset, psi	Ultimate tensile strength (min)	Elongation in 2 inches (min) percent	reduction in area (min) percent	Brinnell hardness (max) 4/
440C 3/	A	Hot	0 500 and less	----	135,000 (max)	----	----	----
			over 0.500	----	----	----	----	269
		Cold	0.500 and less	----	135,000 (max)	----	----	----
			over 0 500	----	----	----	----	285
446	A	Hot	0 500 and less	----	105,000 (max)	----	----	----
			over 0 500	----	----	----	----	223
		Cold	0.500 and less	----	110,000 (max.)	----	----	----
			over 0 500	----	----	----	----	235
UNS S30430	A	Hot	0.500 and less	----	115,000 (max.)	----	----	----
			over 0 500	25,000	65,000	60	70	----
		Cold	0.500 and less	----	115,000 (max.)	----	----	----
			over 0 500	30,000	70,000	65	80	----
384	A	Hot	0.500 and less	----	115,000 (max)	----	----	----
			over 0 500	25,000	70,000	50	75	----
		Cold	0 500 and less	----	115,000 (max)	----	----	----
			over 0 500	25,000	70,000	50	75	----

1/ A tensile strength of 70,000 p s i minimum is acceptable for extruded material.
2/ Material shall be capable of a maximum hardness as follows when oil quenched from 1742°F (950°C): 250-405 HB
3/ Material shall be capable of being heat treated to the following minimum hardnesses·

420 - HRC 50
440A - HRC 54
440B - HRC 56
440C - HRC 58

4/ Or equivalent Rockwell hardness.

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Table IV Cross reference of corrosion resisting steel designations

Federal designations			Military designations		Unified Numbering System
This specification		QQ-S-763A	46S18E	MIL-S-7720	
Class	Condition	Class and type	Class and type	Compostion	
202	---	---	---	---	S20200
302	---	---	---	302	S30200
302	A	1-D	1-B, 1-D	---	---
302	B	1-B, 1-C	1-C	---	---
XM-7 4/	A	---	---	---	S30430
304	---	---	---	---	S30400
304	A	1-A	1-A	---	---
304	B	---	---	---	---
304L	---	---	---	---	S30403
305	---	---	---	---	S30500
309	---	---	---	---	S30900
310	---	---	---	---	S31000
310	A	---	---	---	---
316	---	---	---	316	S31600
316	A	9-A	9-A	---	---
316	B	9-C	9-C	---	---
316L	---	---	---	---	S31603
317	---	---	---	---	S31700
321	---	---	8	---	S32100
321	A	8A	---	---	---
347	---	---	8	---	S34700
347	A	8A	---	---	---
384	A	---	---	---	S38400
403	---	---	---	---	S40300
405	---	---	---	---	S40500
410	---	---	---	---	S41000
410	A	---	3-A	---	---
410	A (Hot fin)	3-A	---	---	---
410	A (Cold fin)	3-C	3-C	---	---
410	T	3-E 1/	3-E	---	---

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Table IV Cross reference of corrosion resisting steel designations (continued)

Federal designations			Military designations		Unified Numbering System
This specification		QQ-S-763A	46S18E	MIL-S-7720	
Class	Condition	Class and type	Class and type	Compostion	
410	H	3-F	3-F	---	---
414	---	---	---	---	S41400
420	---	5-F 2/	---	---	S42000
420	A 3/	5-A	5A, 5F	---	---
430	---	---	4	---	S43000
430	A	4	---	---	---
440A	---	---	---	---	S44002
440B	---	---	---	---	S44003
440C	---	---	---	---	S44004
440C	A	10A	---	---	---
446	---	---	---	---	S44600

1/ Not identical, but is an acceptable substitute.
2/ Shall be heat treated to HRC 50 minimum
3/ This is not in the hardened condition to corresponding to 5F (46S18E) but is required to be capable of being heat treated to HRC 50 minimum with a suitable stress relief Parts should be hardened to the requirements of the drawing or as specified by the procuring activity.
4/ XM-7 is no longer used as the AISI designation The correct designation is the Unified Numbering System designation UNS S30430

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